# Getting Started with OpenLDAP

## Part 3

# Understanding the OpenLDAP Schema

**In this article, we'll begin our journey of understanding LDAP.**

To understand ACLs and LDAP security better, we need to understand the underlying structure of data in a directory. To do this, let us go back to the *ldif* file that we created in the first article of this series, to populate our directory:

```
[vbg@vbg-work openldap]$ cat /tmp/ldap.ldif
dn: dc=knafl,dc=org
dc: knafl
objectClass: top
objectClass: domain

dn: ou=addressbook,dc=knafl,dc=org
ou: addressbook
objectClass: top
objectClass: organizationalUnit

dn:cn=Varad Gupta,ou=addressbook,dc=knafl,dc=org
cn: Varad Gupta
sn: Gupta
l: Gurgaon
street: M-37, Old DLF Colony, Sector-14
st: Haryana
postalCode: 122003
homePhone: 0124 22222222
mobile: 0999999999
mail: varad.gupta@fosteringlinux.com
objectClass: top
objectClass: inetOrgPerson
```

From the above file, the following facts can be deduced:

First, the directory consists of three records or entities:

- The Distinguished Name (DN) of the first record(entity) is dc=knafl,dc=org.
- The other two DNs are *ou=addressbook, dc=knafl, dc=org*, and *cn=Varad Gupta, ou=addressbook, dc=knafl, dc=org.*

Second, there are no duplicate DNs (there cannot be!). Third, the data can be represented in a hierarchical tree format. (We can see that dc=knafl is below dc=org and that ou=addressbook is below dc=knafl and so on.)

We now understand what DN is, but what is 'dc', or 'cn' for that matter? What does 'objectClass' mean or represent?

To understand these terms, we need to look at the structure or schema of the underlying directory. Simply put, a schema is the structure in which data units (and, therefore, records/entities) are organised in a directory. All of us confront schemas on a daily basis. Any spreadsheet that we make consists of rows and columns. The columns define the schema of the spreadsheet. Those of us who are familiar with database management systems create tables and define their schemas regularly.

Similarly, a directory has a schema. The ability to use different schemas (or schemata) for a single record enables a directory to store records with different data—providing much greater flexibility. Going back to our discussion on the *slapd.conf* configuration file, the top few files in the global section define the various schemas that can be used: